WORKED SOLUTION

STEP II 1995, Question 14

Probability and Statistics • Hypothesis Testing

STEP II 1995, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q14

The key idea

Write the density as $x^2\phi(x)$, where ϕ is the standard normal density. Then every integral we need becomes a moment of a standard normal Z, which is where the hint about $\mathrm{E}(Z^4)$ comes in. To decide whether data "shakes our faith", we ask how probable a result at least this extreme would be if our hypothesis ($\mu = 0$) were true. If that probability is tiny, and a shift in μ would make the data unremarkable, the data count against us.

Solution

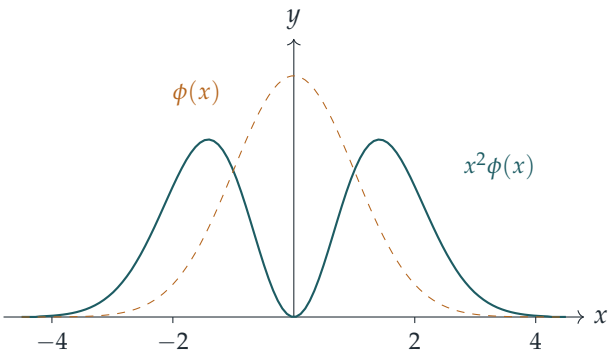
Finding A

Integrating by parts with $u = x$ and $\mathrm{d}v = x\mathrm{e}^{-x^2/2}\,\mathrm{d}x$, so $v = -\mathrm{e}^{-x^2/2}$:

$$\int_{-\infty}^{\infty} x^2\mathrm{e}^{-x^2/2}\,\mathrm{d}x = \left[-x\mathrm{e}^{-x^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\,\mathrm{d}x = 0 + \sqrt{2\pi},$$

using the standard normal integral. The total probability must be 1, so

$$A = \frac{1}{\sqrt{2\pi}}, \qquad \text{that is,} \qquad \mathrm{f}(x) = x^2\phi(x), \quad \phi(x) = \tfrac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}.$$



The density is symmetric with two humps at $x = \pm\sqrt{2}$ and is zero at $x = 0$. By symmetry $\mathrm{E}(X) = 0$, and

$$\mathrm{E}(X^2) = \int_{-\infty}^{\infty} x^4\phi(x)\,\mathrm{d}x = \mathrm{E}(Z^4) = 3, \qquad \text{so} \qquad \mathrm{Var}(X) = 3.$$

Part (i)

Under our hypothesis, how likely is a result as far out as $a = 87.3$? Integrating by parts as before (now $v = -\phi(x)$, since $\phi'(x) = -x\phi(x)$),

$$\mathrm{P}(X \geq a) = \int_a^{\infty} x \cdot x\phi(x)\,\mathrm{d}x = a\phi(a) + \mathrm{P}(Z \geq a).$$

For $a > 0$ we also have $\mathrm{P}(Z \geq a) = \int_a^{\infty} \phi(x)\,\mathrm{d}x \leq \int_a^{\infty} \frac{x}{a}\phi(x)\,\mathrm{d}x = \frac{\phi(a)}{a}$, since $x/a \geq 1$ on the range. Hence, by symmetry,

$$\mathrm{P}(|X| \geq 87.3) \leq 2\left(a + \frac{1}{a}\right)\phi(a) \approx 70\,\mathrm{e}^{-3810.6} \approx 10^{-1653}.$$

This is absurdly small: if our hypothesis were true, we should essentially never see such a value. On the other hand, if μ were around 85 or 89 (so that 87.3 sits on one of the humps of $\mathrm{f}(x - \mu)$), a result of 87.3 would be entirely ordinary. So a single result of 87.3 should shake our faith completely. The only escape is to doubt the result itself (a recording error, say), and a sensible scientist would check that before abandoning the hypothesis.

Ask yourself Isn't it unfair to judge from one observation, when every particular value has probability zero?

The point is not the probability of the exact value 87.3, but of a value *at least this far* from where our hypothesis puts the mass. Equally, the density at 87.3 under our hypothesis is roughly 10^{-1650} times the density there under a suitable alternative. Either way, one observation is ample when it is this extreme.

Part (ii)

Let $X_1, \ldots, X_{1000}$ be the independent results and $\bar{X}$ their mean. Under our hypothesis each X_i has mean 0 and variance 3, so

$$\mathrm{E}(\bar{X}) = 0, \qquad \mathrm{Var}(\bar{X}) = \frac{3}{1000}, \qquad \mathrm{sd}(\bar{X}) = \sqrt{0.003} \approx 0.0548.$$

The variance is finite, and 1000 is large, so by the central limit theorem $\bar{X}$ is approximately $\mathrm{N}(0, 0.003)$, even though f itself looks nothing like a normal density. The observed mean is

$$\frac{0.23}{0.0548} \approx 4.2$$

standard deviations from 0, and

$$\mathrm{P}(|\bar{X}| \geq 0.23) \approx \mathrm{P}(|Z| \geq 4.2) \approx 3 \times 10^{-5}.$$

So if our hypothesis were true, a sample mean this far from 0 would turn up about three times in a hundred thousand repetitions of the whole experiment. Under the alternative with $\mu \approx 0.23$, the same data would be completely typical. This should also shake our faith seriously, though less overwhelmingly than in (i). We would reject $\mu = 0$ at any conventional significance level, e.g. 1% or even 0.1%.

Ask yourself Could we avoid the central limit theorem by using Chebyshev's inequality?

Chebyshev's inequality gives $\mathrm{P}(|\bar{X}| \geq 0.23) \leq 0.003/0.23^2 \approx 0.057$ without any approximation. That is suggestive but not decisive, since results this extreme would then be ruled out only to about 6%. The central limit theorem gives the much sharper figure, and with 1000 observations of a distribution whose tails decay like $\mathrm{e}^{-x^2/2}$, the approximation can be trusted.

Checks

- Simpson's rule gives $\int x^2 \mathrm{e}^{-x^2/2}\,\mathrm{d}x = 2.506628\ldots = \sqrt{2\pi}$, and $\int x^2 \mathrm{f}(x)\,\mathrm{d}x = 3.000$.
- The tail formula $\mathrm{P}(|X| \geq a) = 2(a\phi(a) + \mathrm{P}(Z \geq a))$ agrees with numerical integration at $a = 3$ (both 0.029291).
- $|X|$ has the same density as the length of a vector of three independent $\mathrm{N}(0,1)$ variables. Simulating X this way and averaging 1000 values 4000 times gave about 4.9% of means beyond 2 sd, close to the normal prediction of 4.6%, and none beyond 0.23.

Common slips

- Taking $\mathrm{Var}(X) = 1$ because the exponential looks like a standard normal. The extra x^2 pushes mass outwards, giving variance 3.

- In (ii), applying a normal test to a single X rather than to $\bar{X}$, or forgetting to divide the variance by 1000.
- In (i), arguing that "one observation proves nothing". With an outcome this extreme, it does.

Beyond STEP

This is hypothesis testing seen from first principles. A first-year statistics course formalises it with p-values, likelihood ratios (the Neyman–Pearson lemma) and the central limit theorem. Incidentally, $|X|$ has the chi distribution with 3 degrees of freedom, which is the Maxwell–Boltzmann distribution of molecular speeds in a gas. Grimmett and Welsh, *Probability: An Introduction*, covers the limit theorems used here.

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